

On-Chip Mental Stress Detection: Integrating a Wearable Behind-The-Ear EEG Device With Embedded Tiny Neural Network

Ngoc-Dau Mai  and Wan-Young Chung , Senior Member, IEEE

I. INTRODUCTION

Abstract—The study introduces an innovative approach to efficient mental stress detection by combining electroencephalography (EEG) analysis with on-chip neural networks, taking advantage of EEG's temporal resolution and the computational capabilities of embedded neural networks. The proposed system utilizes behind-the-ear (BTE) EEG signals and on-chip neural networks for mental stress detection. A wearable custom-designed device captures EEG signals from a single BTE channel, performs on-chip signal-to-spectrogram conversion, and integrates a compact convolutional neural network (CNN) for stress classification. The system systematically identifies key EEG frequency bands associated with stress and includes a user-friendly smartphone application for intuitive stress monitoring. EEG data were collected from 15 participants during stress-inducing tasks, such as Stroop and Mental Arithmetic tests. On-chip processing is essential for filtering EEG noise, converting signals into spectrogram images, and using these images as inputs for stress detection through the proposed on-chip CNN model. The experimental results demonstrate strong performance: leave-one-out cross-validation (LOOCV) achieves 91.72% accuracy, 93.74% specificity, 88.69% sensitivity, 90.43% precision, and an F1-score of 0.8955; while 10-fold cross-validation (CV) yields 95.32% accuracy, 95.89% specificity, 94.47% sensitivity, 93.95% precision, and an F1-score of 0.9421 on untrained datasets. The Beta band (13 Hz to 30 Hz) is identified as the most significant frequency band for detecting mental stress. This integration of BTE EEG analysis with on-chip CNNs represents a significant advancement in mental stress detection and has potential applications in medical assistance tools.

Index Terms—Mental stress detection, BTE-EEG, On-Chip processing, tiny neural networks.

Received 23 May 2024; revised 22 July 2024 and 21 October 2024; accepted 10 December 2024. Date of publication 17 December 2024; date of current version 7 March 2025. This work was supported by the National Research Foundation of Korea (NRF) through the Korea government (MSIT) under Grant 2019R1A2C1089139. (Corresponding author: Wan-Young Chung.)

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by Pukyong National University Institutional Review Board under Application No. 1041386-202003-HR-11-02.

The authors are with the Department of AI Convergence, Pukyong National University, Busan 48513, South Korea (e-mail: ngocdau-mai@gmail.com; wychung@pknu.ac.kr).

Digital Object Identifier 10.1109/JBHI.2024.3519600

STRESS is a complex interaction between the body and mind in reaction to perceived mental, emotional, or physical strain [1]. Chronic stress can contribute significantly to the development of various medical conditions, including depression, cardiovascular disease, stroke, and cancer [2], [3], [4]. Managing stress is crucial for preventing severe health complications. Traditionally, stress assessment has relied on methods like standardized questionnaires and analysis of chemical and physiological markers [5], [6], [7]. However, advances in neuroimaging technology have created new pathways for understanding the brain systems that underpin stress reactions. Magnetic Resonance Imaging (MRI), Functional MRI (fMRI), Positron-Emission Tomography (PET), Single-Photon-Emission Computed Tomography (SPECT), and Functional Near-Infrared Spectroscopy (fNIRS) are all techniques that provide valuable insights into how stress affects neural activity in the brain [8], [9], [10], [11], [12]. Despite their potential, these approaches have limitations, particularly the inability to provide continuous, real-time evaluation of an individual's stress levels [13].

The use of physiological reactions is a compelling alternative to established stress-measuring methods. When people are confronted with stressful events, their autonomic nerve systems react, resulting in an imbalance between the sympathetic and parasympathetic systems. This causes physiological changes that may be used to monitor stress levels, such as electroencephalography (EEG), electrodermal activity (EDA), and photoplethysmography (PPG). Among these, EEG stands out as a potent tool for studying stress dynamics. EEG gives trustworthy insights into stress-related aspects by collecting brain electrical activity and transferring it via an amplifier [14], [15], [16]. Stress-induced brain activity patterns are abundantly detectable in EEG recordings [17]. Notably, EEG's capacity to constantly record signals in near real-time enables extensive investigation of stress reactions as they evolve millisecond by millisecond. This makes EEG useful for researching stress in real-world settings, creating online physiological monitoring systems, and integrating passive brain-computer interfaces across several domains. Unlike other physiological responses such as PPG, EDA, and blood pressure, EEG provides rapid feedback without major time delays [18]. However, typical EEG equipment, which is primarily intended for scalp signal recording, presents difficulties

in everyday use due to its bulkiness, design constraints, and lack of mobility [19].

One potential solution to the constraints of typical scalp-based EEG systems is to use EEG data recorded around the ears. Ear EEG electrodes, which are placed either within the ear canal or near the outer ear, have various benefits, including higher wearer mobility, improved skin-to-electrode contact, and increased user comfort [20]. However, this approach may have trade-offs, such as decreased signal amplitude and a restricted focus on the measured brain areas. Ear EEG systems are widely classified into two categories based on electrode placement: in-ear (within the ear canal) and BTE configurations (near and usually behind the earlobe) [19]. Pioneering research by Joong Hoon et al. [21] has explored the use of in-ear EEG, showing promising results in stress detection (achieving 75.8% accuracy). However, the systems used in their studies were non-portable and intrusive for real-world applications due to their reliance on tethered computer interfaces. Moreover, concerns persist regarding the practicality of in-ear EEG designs, which often require sealing the ear canal to ensure electrode contact, potentially affecting wearer awareness of their surroundings and auditory perception. As an alternative, adopting BTE EEG configurations offers a viable solution, overcoming these challenges while maintaining signal quality and user comfort. It's important to note that despite advancements in research, investigations into the development of ear EEG-based stress detection systems are still in their early stages. Therefore, ongoing efforts are needed to fully explore the potential and address the practical limitations of this innovative approach.

The study presents several notable contributions, outlined as follows:

- The study introduces a wearable embedded device for capturing EEG signals behind the ear, enabling integration of miniature neural networks for stress detection directly onto the chip.
- An innovative on-chip signal-to-spectrogram methodology, along with a compact embedded CNN model, is introduced for effectively categorizing various levels of mental stress.
- A user-friendly smartphone application is developed for intuitive monitoring and detection of mental stress, enhancing the practical applicability of the system in real-world scenarios.

The paper's structure entails the following: Section II discusses related work. Section III elucidates the proposed framework underpinning this investigation. Section IV presents the obtained results. Lastly, Section V provides the concluding remarks derived from this study.

II. RELATED WORK

Several studies have investigated the use of behind-the-ear EEG signals in various applications. For instance, Bhagubai et al. [22] employed a wearable Sensor Dot from Byteflies to capture BTE EEG signals from two channels: one across the head and one ipsilateral to the hemisphere of seizure onset. Despite its compact design and the use of fewer channels,

this device necessitated external computing resources for EEG signal processing, thereby complicating the system and limiting portability. Similarly, Ha-Trung et al. [19] proposed a custom-designed embedded device for detecting driver drowsiness, which also utilized BTE EEG signals but required processing from six channels. This configuration increased the device's size, leading to user discomfort, extended signal processing times, higher energy consumption, and reduced operational duration. In contrast, our proposed device addresses these limitations by employing a single-channel BTE EEG signal acquisition system. All signal processing, feature extraction, and neural network classification are conducted directly on-chip. This integrated approach not only conserves energy by eliminating the need for external computing devices but also removes delays associated with data transmission, thereby enhancing the efficiency of handling continuous EEG signal streams.

A significant challenge in implementing current stress detection systems is the conventional setup of wearable EEG-based systems, typically consisting of two main components: a wearable device for EEG signal acquisition and computer-based hardware, like a PC, smartphone, or smartwatch, for data processing [23], [24], [25]. The wearable device captures the raw EEG signal and wirelessly transmits it to the computer-based hardware, which then carries out subsequent stages, such as preprocessing, feature extraction, and stress detection. However, wireless systems encounter issues like increased power consumption and propagation delay with longer transmitted packets, resulting in significant delays in the overall system. Moreover, relying on external computing devices for processing EEG signals further complicates the system, making it cumbersome and less portable. To tackle these challenges, one approach is to move all computational stages to the wearable EEG device itself, a concept known as on-device processing. This strategy separates computation from wireless communication latency and reduces or eliminates the power consumption associated with wireless communication.

In the implementation of on-device processing, a crucial step is deploying stress detection models. To accomplish this, we have embraced Tiny Machine Learning (TinyML) [26], a framework designed to optimize models for efficient operation on microcontroller unit (MCU) platforms. TinyML has emerged as a promising solution that bridges the gap between microcontrollers and machine learning (ML), especially deep learning. Depending on the system architecture, TinyML offers several key advantages. Firstly, in terms of privacy, processing data at the edge device rather than transmitting it to a central device (such as the cloud) can help mitigate potential privacy breaches or interception vulnerabilities. Secondly, concerning latency, since all processing occurs on the edge device, concerns about wireless latency are bypassed, enabling real-time decision-making tasks at the edge. Lastly, regarding energy efficiency, while the processing stage may be energy-intensive, the power consumption associated with data transmission can often exceed it. By consolidating processing on the edge device, overall energy consumption is significantly reduced.

Most EEG signal collection devices used in prior research, which transmit raw EEG signals to a PC for processing, exhibit

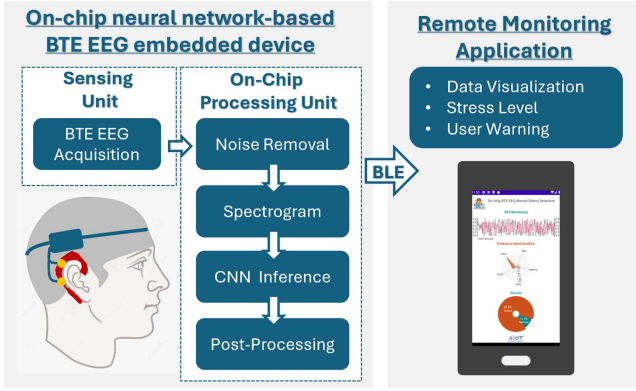


Fig. 1. The overview of the end-to-end on-chip integrated wearable BTE EEG system designed for real-time detection of mental stress.

high power consumption. For instance, S. Chaabene et al. [27] employed the Emotiv EPOC+ to collect scalp EEG signals from 14 channels. This device, powered by a 3.7 V, 640 mAh battery, consumed 395 mW and had a maximum operational time of 6 hours. Similarly, S. Gangadharan K et al. [28] used the Muse-2 device for collecting EEG signals from 4 temporal channels. This device, which relies on a 3.7 V, 290 mAh battery, consumed 214 mW and lasted up to 5 hours. Furthermore, N. Pham et al. [29] utilized a custom-designed device to collect EEG signals from 8 channels located behind the ear. This device, powered by a 3.7 V, 600 mAh battery, consumed 241.3 mW and provided up to 9.2 hours of operation. In contrast, our study evaluated a device with a 3.7 V, 500 mAh battery, measuring total power consumption and battery life. Continuously operating without sleep, our device consumes approximately 66.254 mW and offers around 27.92 hours of battery life. These findings, detailed in the “IV. Results and Discussion” section, demonstrate that our device significantly outperforms previous EEG devices in terms of energy efficiency and operational duration.

III. MATERIALS AND METHODS

A. Framework Description

In this study, we propose an end-to-end on-chip integrated wearable BTE EEG system for real-time mental stress detection, consisting of two main components as illustrated in Fig. 1: an on-chip neural network-based BTE EEG embedded device and a remote monitoring application. The first component, the on-chip neural network-based BTE EEG embedded device, includes two primary units. The sensing unit collects EEG signals from an earpiece sensor electrode positioned behind the ear, which are then filtered through an RC filter to remove interference, amplified, and converted into digital data by an analog-to-digital converter for further processing on the MCU. The second unit, the on-chip processing unit, is pivotal in the system by eliminating EEG noise, converting signals into spectrogram images through on-chip signal-to-spectrogram conversion, and utilizing these images as input for the on-chip CNN model to detect stress levels. Subsequently, essential information undergoes on-chip

post-processing before being transmitted to the remote monitoring application. This user-friendly smartphone application enables data visualization, displays detected stress levels, and promptly alerts users to stress states.

B. On-Chip Processing BTE Embedded Device

Our on-chip processing embedded device consists of four units: the Earpiece unit, Processing unit, Communication unit, and Power supply unit (shown in Fig. 2). The Earpiece unit, 3D-printed with flexible material for wearer comfort, houses EEG electrodes positioned according to cEEGrid [30] standards (R3 and Ref positions), with dry-contact gold cups ensuring signal consistency. Anti-aliasing filters ($R = 4.99 \text{ k}\Omega$ and $C = 4.7 \text{ nF}$) prevent aliasing by allowing frequencies below the cutoff frequency to pass through. The INA118 serves as a precision instrumentation amplifier, accurately amplifying small brain signals while rejecting noise. The processed EEG signal is then transmitted to the Processing unit, starting with the ADS1115, a precision, low-power, 16-bit analog-to-digital converter (ADC) commonly used in medical applications due to its high data rate capabilities (up to 860 SPS) and wide input ranges ($\pm 256 \text{ mV}$ to $\pm 6.144 \text{ V}$). The ADS1115 includes an input multiplexer (MUX) for multiple input measurements. The main function is managed by the ESP32S3 processor, a dual-core, 32-bit processor operating at up to 240 MHz, interfacing with the ADS1115 via I2C protocol. The EEG data utilized in this study were sampled at a rate of 128 Hz. Upon receiving data, initial on-chip noise removal is performed using a band-pass filter (1 Hz to 60 Hz). The filtered signal is then converted into a spectrogram image using the Fast Fourier Transform (FFT) for input to the on-chip CNN model for stress classification. Post-processing is conducted before transmitting results to a smartphone application through Bluetooth Low Energy (BLE) in the Communication unit. The Power supply unit comprises the MCP73831 battery charge management controller and a Li-ion Polymer Battery (3.7 V, 500 mAh) to power the device.

C. BTE EEG Signal-to-Spectrogram Conversion

The Discrete Fourier Transform (DFT) [31] converts a finite sequence of equally spaced function samples into another sequence of the same length, representing their frequency domain counterparts. In mathematical terms, the DFT of a given sequence x_n with n ranging from 0 to $N - 1$ is defined as follows:

$$X_k = \sum_{n=0}^{N-1} x_n e^{-\frac{2\pi i}{N} nk}, \quad 0 \leq k \leq N - 1 \quad (1)$$

Where X_k is the complex-valued result of the DFT at frequency index k , x_n represents the input sequence, and N is the total number of samples in the sequence. The term $e^{-\frac{2\pi i}{N} nk}$ denotes the complex exponential function, and i represents the imaginary unit.

The FFT [32] technique efficiently computes the DFT with a computational complexity of $O(n \log n)$, significantly reducing

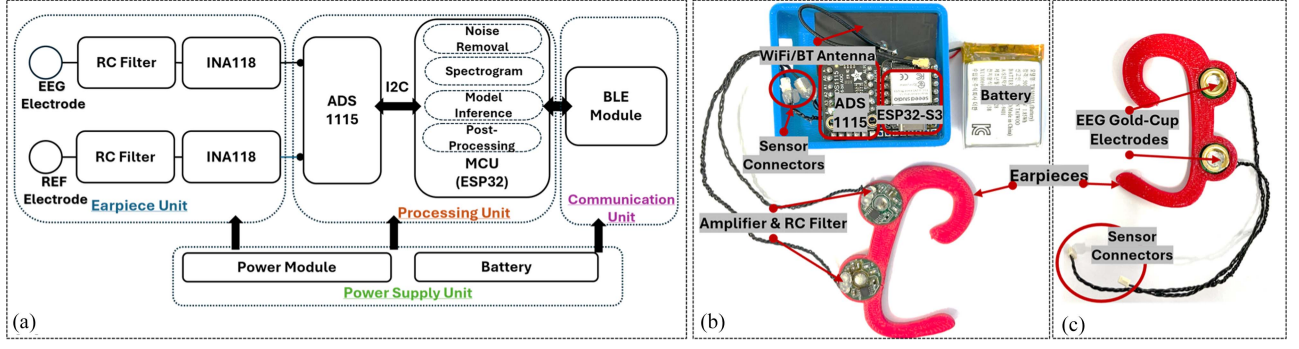


Fig. 2. (a) The block diagram detailing the proposed system on a chip, (b) the overview of the proposed device, and (c) the earpiece unit featuring gold cup sensors.

TABLE I
BTE-EEG SIGNAL-TO-SPECTROGRAM CONVERSION PROCESS

BTE-EEG Signal2Spectrogram Conversion	
Input: EEG signal as $x(t)$, where t presents time.	
Output: Spectrogram matrix $S(f, t)$, where f is the frequency axis and t is the time axis.	
Step 1	Divide the EEG signal into segments: Divide $x(t)$ into overlapping segments, denoted as $x_i(t)$, with each segment having a length L and an overlap O . Here, i ranges from 1 to N , where N represents the total number of segments.
Step 2	Apply a window function: Apply a window function, $w(t)$, to each segment, $x_i(t)$, for multiplication. $x_{w,i}(t) = x_i(t) \cdot w(t)$
Step 3	Compute the DFT: Use the FFT on each windowed segment, $x_{w,i}(t)$, to extract its frequency content. $X_{w,i}(f) = FFT(x_{w,i}(t))$
Step 4	Calculate the power spectrum: Compute the squared magnitude of the FFT output to derive the power spectral density for each segment. $P_i(f) = X_{w,i}(f) ^2$
Step 5	Repeat for all segments: Slide the window along the EEG signal with the specified overlap O , repeating steps 2 to 4 for each segment.
Step 6	Aggregate results: Stack the power spectra of each segment along the time axis to create the spectrogram matrix, $S(f, t)$.

it from the original $O(n^2)$. This efficiency is achieved by breaking down the DFT matrix into sparse factors, most of which are zero.

In the study, we employ FFT to analyze EEG signals, enabling accurate computation of frequency components and differentiation between various cognitive states and neurological diseases. Although techniques like Continuous Wavelet Transform (CWT) and Superlet Transform (SLT) offer superior time localization, FFT remains the most efficient and fastest method for signal processing [33]. Its rapid performance makes it ideal for converting EEG signals into spectrograms directly on-chip. Additionally, to generate spectrogram images from BTE EEG signals, we introduce an on-chip FFT-based BTE EEG spectrogram conversion. This method transforms time-varying EEG signals into 2D matrices with time and frequency axes using spectrograms, facilitating analysis in the time-frequency domain. In spectrogram images, brighter areas represent higher power or the presence of components, while darker areas indicate lower power or the absence of components. Table I provides

a comprehensive explanation of the BTE EEG spectrogram conversion process.

D. Convolutional Neural Network-Based Stress Detection

In this study, we utilized a CNN [34] model based on Deep Learning (DL) [35] to classify the obtained spectrogram images. CNNs are renowned for their ability to effectively classify data and discover relevant features autonomously. With multiple convolution layers, CNN architectures can learn high-level features and semantic representations, as well as low-level details such as edges and textures. This hierarchical learning aids CNNs in comprehending complex visual patterns.

A fundamental aspect of CNNs is the convolution operation, which is crucial for extracting and recognizing important characteristics in images, enabling the network to perform tasks like object detection and image classification. The internal mechanisms of the convolutional layer, characterized by a spatial filter dimension of $P \times Q$ and C channels, can be described as follows:

$$Y_{m, n, k} = ACT \left(\sum_{p=1}^P \sum_{q=1}^Q \sum_{c=1}^C W_{p,q,c} \cdot X_{m+p-1, n+q-1, c} + b_k \right) \quad (2)$$

In the provided formula, $Y_{m, n, k}$ represents the value of the output feature map at the m_{th} row, n_{th} column, and k_{th} channel. $X_{m+p-1, n+q-1, c}$ denotes the value at the input feature map's $(m+p-1)_{th}$ row, $(n+q-1)_{th}$ column, and c_{th} channel. p_{th} row, q_{th} column, and c_{th} channel of the filter for the k_{th} channel of the output feature map is denoted by $W_{p,q,c}$. The bias term for the k_{th} channel of the output feature map is denoted by b_k . The element-wise sum's activation function is indicated by the function $ACT(\cdot)$. This formula enables the calculation of the dot product of the filter weights W and the corresponding area of the input feature map X . The result is then summed with the bias term b and the activation function $ACT(\cdot)$ is applied to introduce non-linearity to the output.

In this study, we propose a CNN model using BTE EEG images, created from converting signals to spectrograms, as inputs

TABLE II
HYPERPARAMETER TUNING RESULTS FOR THE PROPOSED CNN MODEL

Hyper-parameters/Layers	Search Space	Optimal Value	Operation	*AF	Output Shape	Param#
<i>Input</i>	(112x112x1)	(112x112x1)	---	---	---	---
<i>Conv2D</i>	[32, 16, 8]	16	k=3, s=1, p=1	ReLU	(-1, 16, 112, 112)	160
<i>MaxPooling2D</i>	---	---	k=2, s=2	---	---	---
<i>Conv2D</i>	[32, 16, 8]	32	k=3, s=1, p=1	ReLU	(-1, 32, 56, 56)	4,640
<i>MaxPooling2D</i>	---	---	k=2, s=2	---	---	---
<i>Conv2D</i>	64	64	k=3, s=1, p=1	ReLU	(-1, 64, 28, 28)	4,640
<i>MaxPooling2D</i>	---	---	k=2, s=2	---	---	---
<i>Flatten</i>	---	---	---	---	---	---
<i>Dense</i>	128	128	---	ReLU	(-1, 128)	1,605,760
<i>Dense</i>	64	64	---	ReLU	(-1, 64)	8,256
<i>Output</i>	2	2	---	Softmax	(-1, 2)	8,256
<i>Optimizer</i>	[Adam, SGD]	Adam	---	---	---	---
<i>Learning Rate</i>	[0.01, 0.001, 0.0001]	0.001	---	---	---	---
<i>Batch Size</i>	[4, 8, 16, 32]	16	---	---	---	---
<i>Epoch</i>	200	200	---	---	---	---

*AF: activation function, **k**: kernel size, **s**: stride, **p**: padding.

to detect stress levels, categorizing outputs into two classes: non-stress and stress states. The architecture of the proposed CNN model is presented in Table II, where hyperparameters have been fine-tuned to optimize the model's performance in stress detection. The detailed process of hyperparameter tuning for the model is discussed in Section IV-C. Data Preparation and Training Strategy. The CNN architecture for stress classification comprises three convolutional layers activated by the Rectified Linear Unit (ReLU) function, followed by max-pooling layers to condense features. The convolutional layers utilize 3x3 kernel sizes and produce 16, 32, and 64 feature maps successively. Max-pooling layers are employed to reduce spatial dimensions. Subsequently, the flattened feature maps pass through two fully connected layers consisting of 128 and 64 neurons each, facilitating abstraction. Finally, the output layer consists of two neurons corresponding to stress and non-stress classes for stress level classification. During training, the Adam optimizer is utilized with a learning rate of 0.001 to minimize categorical cross-entropy loss. The proposed classification model functions in a continuous loop: EEG signals are collected and filtered every second, with each segment consisting of 128 data points, to eliminate noise and produce clean signals. These filtered signals are subsequently converted into spectrogram images using FFT-based transformation. These spectrogram images are then input into an embedded CNN model, which performs real-time stress level classification directly on the device.

E. Performance Evaluation Techniques

In our study, we utilized two validation approaches, namely leave-one (subject)-out cross-validation (LOOCV) and k-fold cross-validation (CV), to evaluate the performance of our model. In k-fold CV, the data is divided into k equal parts for training and validation, with results averaged across iterations. Specifically, we employed a 10-fold CV in this study. In LOOCV, each subject's data is used for training on all subjects except one, which is reserved for validation. This process is repeated for each subject, ensuring that each subject's data is used for validation exactly once. Performance metrics include accuracy (Acc), specificity

TABLE III
EVALUATION METRICS USED IN THE STUDY

Metrics	Formula
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$
Specificity	$\frac{TN}{TN + FP}$
Sensitivity	$\frac{TP}{TP + FN}$
Precision	$\frac{TP}{TP + FP}$
F1-Score	$2 \cdot \frac{\text{Precision} \cdot \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}}$

Where *TP*, *TN*, *FP*, and *FN* represent the counts for true positives, true negatives, false positives, and false negatives, respectively.

(Spe), sensitivity (Sen), precision (Pre), and F1 score (F1), which provide insights into the model's classification performance (refer to Table III). Accuracy gauges overall correctness, specificity measures correct identification of negatives, sensitivity assesses correct identification of positives, and precision evaluates the proportion of correctly identified positives among all positives. F1 score balances precision and recall, offering a comprehensive evaluation of classifier performance. To visualize the classifier's reliability, Receiver Operating Characteristic (ROC) graphs are employed. The area under the curve (AUC) in ROC graphs indicates the discriminating capacity of the classifier. An AUC of 1 represents perfect discrimination, while ≤ 0.5 denotes no discrimination. These metrics offer valuable information on the classifier's performance on test data, aiding in the assessment of its effectiveness.

Our proposed model's decisions were interpreted using the Gradient-weighted Class Activation Mapping (GRAD-CAM) visualization technique [36], which employs color maps overlaid on input images to highlight important areas for the predicted class. Red areas indicate spatial features relevant to the predicted class, helping users understand the model's behavior by emphasizing key image components influencing classification decisions and ensuring decisions are based on pertinent features.

F. On-Chip Deployment

We perform EEG signal collection, noise removal, signal-to-spectrogram conversion, and model inference directly on our embedded device (on-chip). The on-chip deployment process is elaborately depicted in Fig. 5. This strategy guarantees a compact, wearable, and portable system, thereby minimizing delays and data loss in contrast to conventional methods involving communication between data acquisition devices and computers. By obviating the necessity to transmit data between the acquisition device and the computer, we reduce complexity and optimize system performance.

We integrated our pre-trained models into the embedded device using TensorFlow Lite for Microcontrollers (TFLM), optimized for running TensorFlow-based NN models on MCUs with minimal memory usage. TFLM converts trained models into a compact FlatBuffer (FB) format, comprising 32-bit floating-point weights, which can be translated into C/C++ code and deployed on embedded devices. Quantization, using the min-max approach, reduces model size and improves hardware latency with minimal degradation of model accuracy by converting weights from 32-bit floating-point precision to 8-bit integer precision [37].

Min-max quantization optimizes the models for deployment on devices with limited resources by mapping weights and activation tensors' floating-point values to lower-precision integers. It uses the tensors' minimum (min) and maximum (max) values to compute a scale factor (s) and zero point (z), given by $s = \frac{\max - \min}{2^b - 1}$ and $z = \lfloor \frac{-\min}{s} \rfloor$ where b denotes the bit-width of the quantized format (e.g., 8 bits for int8). Each tensor value (x) is quantized to $q = \lfloor \frac{x}{s} \rfloor + z$, and during inference, these integers are dequantized to approximate floating-point values using $x' = s \cdot (q - z)$. The resulting C/C++ FB code, along with the TFLM library, is stored in MCU memory. For model updates, only the FB-converted weights are replaced, simplifying the process by delivering them via BLE from the Android app to the embedded device.

G. Remote Monitoring Application

A smartphone application was developed to enable interaction with the BTE EEG-embedded device via BLE communication. Designed using Android Studio and Java, the application supports devices running Android 7.0 (API 24) and higher, ensuring broad compatibility. Key features of the application include device management, real-time visualization of EEG data through a line chart, detailed frequency analysis via a radar chart, and the display of predicted stress states with a pie chart showing percentages for both non-stress and stress conditions. Once the embedded device completes processing, data is seamlessly transmitted to the smartphone application, maintaining a continuous flow of information. To ensure secure BLE data transmission, a password verification mechanism is employed. This system requires that the input from the smartphone match a predefined password before allowing data transmission, thereby limiting access to authorized devices and ensuring data integrity and confidentiality. Fig. 10 illustrates the user interface (UI) of

the smartphone application, showcasing its design and functionalities.

IV. RESULTS AND DISCUSSION

A. BTE EEG Signal Quality Evaluation

To assess the quality of EEG signals collected from our embedded device, we conducted a "Berger effect" test, demonstrating a significant increase in EEG oscillation amplitude in the alpha band (8-13 Hz) when individuals close their eyes [38]. Participants engaged in two 2-second resting-state activities, transitioning from eyes-closed to eyes-open conditions. Analysis of power spectral density using the Welch technique revealed a substantial rise in alpha-band power (around 10 Hz) during the eyes-closed state, contrasting with minimal changes in the eyes-open condition (shown in Fig. 6).

B. Experimental Protocol

For our preliminary investigation on EEG measurements during periods of stress, we enlisted 15 healthy participants (with an average age of 30 ± 7 years), all of whom were students at Pukyong National University. Each participant completed two established stress-inducing tasks: the Stroop color and word task (Stroop task) and Mental arithmetic [39]. The Stroop task and mental arithmetic were selected as stress-inducing tasks due to their consistent reliability in eliciting both physiological and psychological stress within a controlled environment. These tasks are straightforward to standardize and implement, and their extensive use in previous studies enhances the credibility of new research findings [39], [40], [41]. Each stress-inducing task comprised sessions involving 3 minutes of relaxation followed by 3 minutes of stress-inducing activity and 5 minutes allocated for evaluation immediately after completing each stress-inducing task, as illustrated in Fig. 3. This process was repeated 5 times, corresponding to 5 experimental trials for each participant. Continuous BTE EEG data were collected using our proposed embedded device throughout the experiments. Following each session, participants filled out the NASA Task Load Index (NASA-TLX) questionnaire [42] and rated their stress levels for the task on a scale of 0 to 20, providing an additional qualitative stress index. This methodology ensured effective stress induction. We primarily relied on the average scores from participants' self-assessment of stress levels and frustration scores from the NASA-TLX questionnaire to assess perceived stress levels. A low score (0-10 points) indicated a non-stress state, while a high score (11-20 points) indicated a state of stress. Findings revealed that the Stroop task induced lower stress levels compared to the Mental arithmetic task (refer to Fig. 4). The experiment was given ethical approval by Pukyong National University Institutional Review Board, reference number 1041386-202003-HR-11-02.

C. Data Preparation and Training Strategy

The EEG signals collected from behind the ear are divided into 1-second segments, each consisting of 128 data points. These segments are then processed using a fourth-order Butterworth

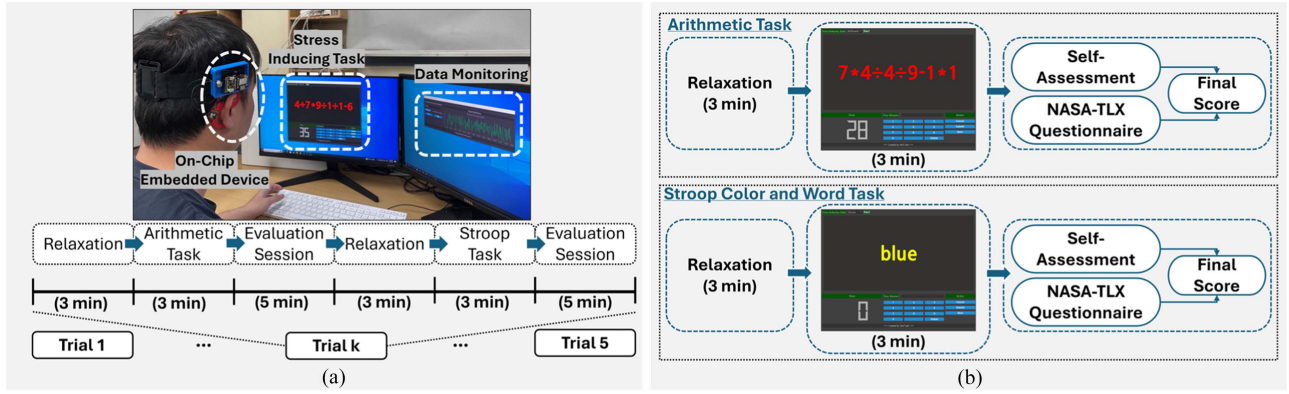


Fig. 3. (a) The experiment setup, (b) The experiment protocol.

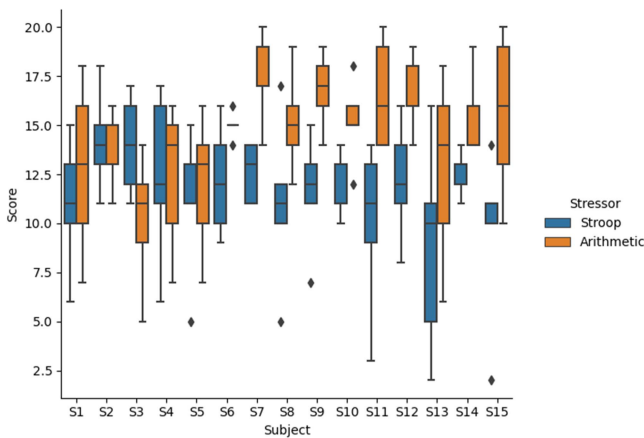


Fig. 4. The score results obtained from the Stroop and Mental Arithmetic tasks.

bandpass filter, designed to eliminate unwanted noise. This filtering process ensures that the resulting EEG signals are clean and confined to the frequency range of 1 Hz to 60 Hz. Subsequently, we employ FFT-based signal-to-spectrogram conversion to create spectrogram images from these EEG segments. These images undergo normalization of pixel intensities within the standard range of 0–255 and are resized to a standardized size of 112×112 pixels, resulting in datasets comprising 32,400 images for stress and 21,600 images for non-stress states.

For evaluation, both 10-fold CV and LOOCV are applied to the datasets. Training of the CNN model is conducted using Python 3.10, leveraging the Keras open-source library and TensorFlow backend, primarily on a high-performance computer equipped with an NVIDIA GeForce RTX3050 graphics card and an 11th Gen Intel(R) Core (TM) i7-11370 processor operating at 3.30 GHz.

Hyperparameter tuning is pivotal in enhancing model training efficacy through the identification of optimal configuration settings. Our study employs the Tree-structured Parzen Estimator (TPE) tuner, a sequential model-based optimization (SMBO) algorithm for Hyperparameter Optimization. TPE tunes $P(x|y)$ and $P(y)$, where x denotes hyperparameters and y signifies the evaluation result. $P(x|y)$ is modeled by reshaping

TABLE IV
CLASSIFICATION PERFORMANCE RESULTS OF CNN AND ANN WITH 10-FOLD CV AND LOOCV

Model	Strategy	Evaluation measures				
		Acc	Spe	Sen	Pre	F1
CNN	10-fold CV	0.9532 ± 0.0020	0.9589 ± 0.0031	0.9447 ± 0.0029	0.9395 ± 0.0043	0.9421 ± 0.0025
	LOOCV	0.9172 ± 0.0032	0.9374 ± 0.0046	0.8869 ± 0.0029	0.9043 ± 0.0065	0.8955 ± 0.0038
ANN	10-fold CV	0.9037 ± 0.0023	0.9277 ± 0.0031	0.8683 ± 0.0042	0.8902 ± 0.0041	0.8712 ± 0.0015
	LOOCV	0.8521 ± 0.0078	0.8850 ± 0.0121	0.8029 ± 0.0067	0.8234 ± 0.0153	0.8129 ± 0.0086

the generative process of hyperparameters, substituting the distributions of the configuration prior with non-parametric densities. Leveraging Microsoft Research's AutoML toolkit, Neural Network Intelligence (NNI) [43], we define the search space, optimization algorithm, and evaluation metric. Please consult Table II for an exhaustive breakdown of the specific hyperparameters utilized in our CNN model.

D. Results

In this study, we developed a comprehensive system for stress detection utilizing EEG signals acquired from behind the ear. Following the removal of unwanted signals, the BTE EEG signals underwent segmentation into 1-second intervals and were then transformed into spectrogram images via FFT-based signal-to-spectrogram conversion. These images served as inputs for our proposed CNN model, tasked with classifying stress states (Stress and Non-stress). We proceeded to train and assessed the CNN model using both 10-fold CV and LOOCV, with the optimal architecture outlined in Table IV. Furthermore, we conducted a performance comparison between our CNN model and a proposed ANN, which retained solely the fully connected layers of the CNN architecture. The experimental results on the testing dataset, detailed in Table IV, highlight the efficacy of our proposed methodology.

The results presented in Table IV indicate that the CNN model consistently outperforms the ANN model in stress classification, regardless of the cross-validation technique used. The CNN

TABLE V
CLASSIFICATION PERFORMANCE OF THE PROPOSED CNN UTILIZING LOOCV ACROSS EACH FREQUENCY BAND

Frequency Band	Acc	Spe	Sen	Pre	F1
Delta	0.8276 ± 0.0174	0.8514 ± 0.0320	0.7918 ± 0.0437	0.7824 ± 0.0358	0.7858 ± 0.0221
Theta	0.8188 ± 0.0016	0.8522 ± 0.0024	0.7688 ± 0.0018	0.7762 ± 0.0029	0.7725 ± 0.0018
Alpha	0.8614 ± 0.0014	0.8689 ± 0.0011	0.8503 ± 0.0036	0.8122 ± 0.0012	0.8308 ± 0.0019
Beta	0.8786 ± 0.0025	0.8775 ± 0.0028	0.8802 ± 0.0044	0.8273 ± 0.0034	0.8529 ± 0.0030
Gamma	0.8527 ± 0.0161	0.8632 ± 0.0251	0.8370 ± 0.0123	0.8042 ± 0.0313	0.8200 ± 0.0176
Full	0.9172 ± 0.0032	0.9374 ± 0.0046	0.8869 ± 0.0029	0.9043 ± 0.0065	0.8955 ± 0.0038

model demonstrates superior performance across all metrics, including accuracy, specificity, sensitivity, precision, and F1-score. Specifically, with 10-fold CV, the CNN model achieves an accuracy of 0.9532 ± 0.0020 , significantly higher than the ANN model's accuracy of 0.9037 ± 0.0023 . Similarly, using LOOCV, the CNN model attains an accuracy of 0.9172 ± 0.0032 , compared to 0.8521 ± 0.0078 for the ANN model. It is noteworthy that the evaluation metrics for both models are lower with LOOCV than with 10-fold CV, mainly due to the substantial impact of the selected testing subject on LOOCV results.

In this study, we investigated the effectiveness of our CNN model in stress detection across various EEG frequency bands. Spectrogram images were generated for five frequency bands: delta (1-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), beta (13-30 Hz), and gamma (30-60 Hz) during the signal-to-spectrogram conversion process. The performance of the CNN model in stress detection across these bands was assessed using LOOCV, a method suitable for real-world applications. The detailed results of this analysis are presented in Table V. Our findings indicate that the CNN model achieved the highest performance when utilizing the full EEG signal band across all five-evaluation metrics. Additionally, the Beta band showed promising results, closely trailing the full band in accuracy and other evaluation metrics, with an accuracy of 0.8786 ± 0.0025 , specificity of 0.8775 ± 0.0028 , sensitivity of 0.8802 ± 0.0044 , precision of 0.8273 ± 0.0034 , and F1-score of 0.8529 ± 0.0030 . Conversely, the Theta band exhibited the lowest performance in stress detection, suggesting its limited suitability for this purpose, with an accuracy of 0.8188 ± 0.0016 , specificity of 0.8522 ± 0.0024 , sensitivity of 0.7688 ± 0.0018 , precision of 0.7762 ± 0.0029 , and F1-score of 0.7725 ± 0.0018 . Notably, the Beta band demonstrated a noteworthy performance, approaching the accuracy levels achieved with the full band. These results underscore the potential of specific EEG frequency bands, particularly the Beta band, in effectively detecting stress using our proposed CNN model.

The ROC curve for the optimal outcomes of each frequency band in LOOCV was plotted, and the results are presented in Fig. 7 for further comparison. As depicted in Fig. 7, the ROC curve of the entire frequency band, with an AUC of 0.9191, closely approaches the (0, 1) point. This suggests that the

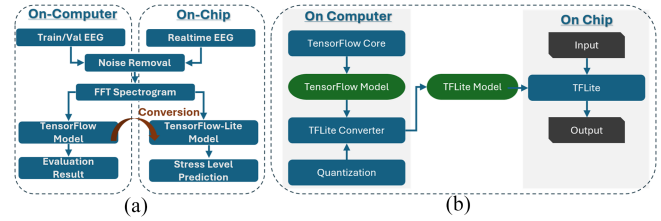


Fig. 5. (a) The on-chip deployment process and (b) the model conversion procedure for on-chip model deployment.

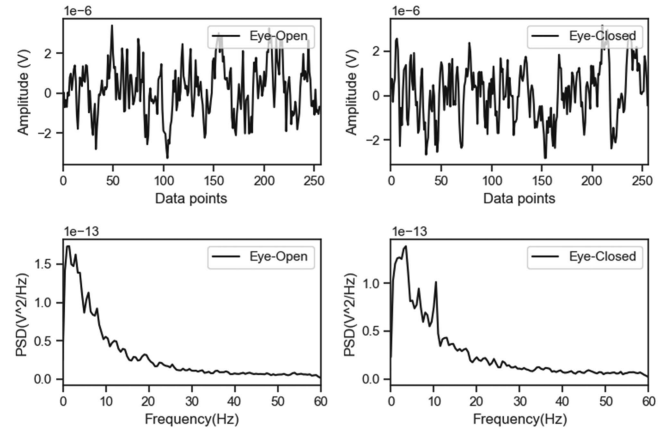


Fig. 6. The evaluation results of BTE EEG quality using the "Berger effect" technique under both eyes-closed and eyes-open conditions.

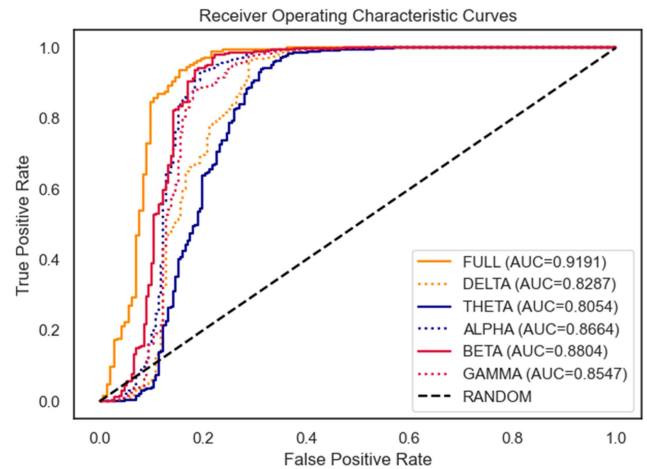


Fig. 7. The comparison of ROC curves for each frequency sub-band.

spectrogram image generated using the 1–60 Hz bandwidth with the trained CNN model serves as a reliable framework for stress detection. Furthermore, among the evaluated frequency sub-bands, the ROC curve and AUC (0.8804) of the beta (13–30 Hz) sub-band closely resemble those of the entire band, with the alpha (8–13 Hz) sub-band's curve and AUC (0.8664) following suit. Conversely, consistent with the evaluation metrics assessment, the theta (4–8 Hz) band exhibited the lowest performance in stress detection, with an AUC of 0.8054, suggesting its limited suitability for this purpose.

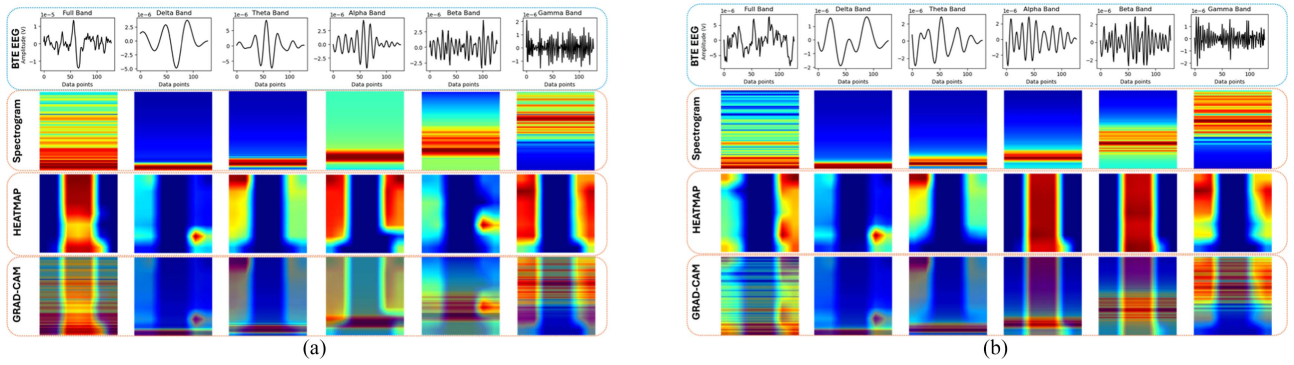


Fig. 8. Heat maps illustrating the Grad-CAM results for both (a) stress and (b) non-stress states across different sub-frequency bands.

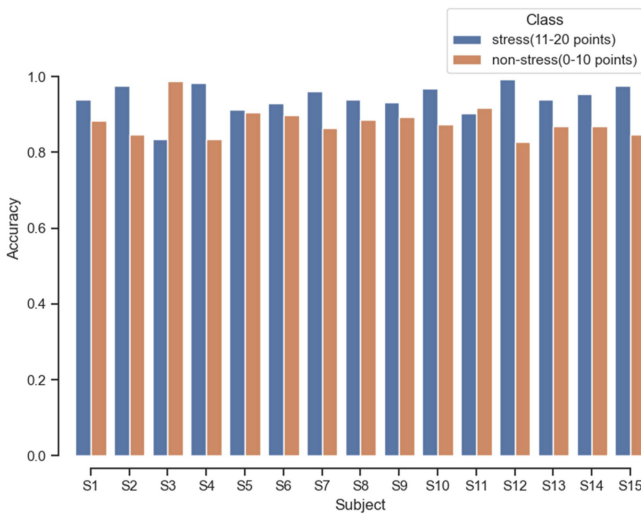


Fig. 9. Accuracy of the proposed CNN model in classifying non-stress states (0-10 points) and stress states (11-20 points) for each individual subject.

In this study, we employed Grad-CAM to enhance the transparency and interpretability of our CNN model utilized for stress classification across spectrogram images derived from five distinct frequency bands. The Grad-CAM results (as shown in Fig. 8), depicted as heat maps corresponding to each frequency band, highlight specific regions where the discriminating patterns between stress and non-stress states vary.

Fig. 9 depicts the accuracy of the proposed CNN model in classifying non-stress states (0-10 points) and stress states (11-20 points) for each individual subject. In this study, each subject's data is utilized as the testing set, while data from the remaining subjects serve as the training set within a LOOCV framework. The results reveal that the accuracy of the CNN model varies across subjects and between the two stress states (stress and non-stress). Notably, the model generally shows higher accuracy for stress state data compared to non-stress state data across most subjects, with the exceptions of subjects S3 and S11. These results indicate that the proposed model demonstrates a particular sensitivity to stress state data, enhancing

its capability to effectively identify when a subject is under stress.

E. End-To-End On-Chip Processing

After a comprehensive comparison of classification results, it becomes evident that our proposed CNN model, utilizing spectrogram images generated from BTE EEG signals across the entire frequency band (1 Hz to 60 Hz) via FFT-based signal-to-spectrogram conversion, demonstrates outstanding performance. Consequently, for chip deployment, we chose to deploy the CNN model trained on BCI EEG spectrogram images covering the entire frequency spectrum using LOOCV, which is well-suited for real-world applications. To facilitate the deployment of the CNN model on a chip, it is necessary to convert it into a format suitable for memory-constrained devices, employing optimizations such as quantization to reduce model size and latency without compromising accuracy, as illustrated in Fig. 5. Initially developed and trained using TensorFlow, the model relies on TensorFlow's interpreter for data processing and output generation. However, TensorFlow's interpreter is tailored for high-performance computers rather than microcontrollers. To address this limitation, TensorFlow Lite serves as a robust interpreter for low-powered devices, with TensorFlow Lite Converter and TensorFlow Lite Interpreter being its key components. The TensorFlow Lite Converter adapts the TensorFlow model to accommodate memory-limited devices, employing techniques like quantization to reduce size and latency while maintaining accuracy. Subsequently, the TensorFlow Lite Interpreter executes the on-device TensorFlow Lite model for predictions based on input data.

Table VI presents the running times for on-chip processing tasks and compares the accuracy of the proposed CNN model before and after conversion to TensorFlow Lite, including quantization, using the same testing set. The findings reveal that the accuracy of both the original CNN model (TensorFlow model) and the converted CNN model (TensorFlow Lite model) remains unchanged. However, there is a slight decrease in accuracy for stress classification after applying quantization optimization, with a reduction of approximately 0.76%, dropping from

TABLE VI

SUMMARY OF THE DURATION FOR EACH ON-CHIP PROCESSING TASK

On-chip Tasks	Duration		Energy Cost (uJ)	Accuracy
	in time (ms)	in MCU Cycles (kCycles)		
Noise Removal	20.12	1609.6	331.38	---
Spectrogram Conversion	402.38	32190.4	6650.77	---
Model Inference	Converted CNN without Quantization	36.54	2923.2	603.21
	Converted CNN with Quantization	31.79	2543.2	524.17
	Post-Processing	2.84	227.2	46.86

TABLE VII

PROPOSED CNN MODEL SIZE IN TERMS OF RAM (kB) AND FLASH (kB) USING TFLM

Memory	Non-Quantized model size (kB)	Quantized model size (kB)
RAM	10.325	8.647
Flash	96.512	43.654

94.32% to 93.56%. This decline is attributed to the conversion process, where weights shift from 32-bit floating-point values to 8-bit integers, thereby altering their precision. Despite the minor reduction in accuracy post-quantization optimization, the model's inference speed significantly improves from 36.54 ms to 31.79 ms.

Power consumption and inference latency are crucial factors for deep-learning models, particularly in battery-constrained embedded applications [44]. Quantized models can significantly reduce both parameters. To evaluate this, we employed the Power Profiler Kit 2 (PPK2) from Nordic Semiconductor, in conjunction with the Power Profiler app (PPA), to measure and analyze on-device computational latency and energy consumption. Table VI presents the duration in time (ms), MCU cycles (kCycles), and energy cost (uJ) for various on-chip processing tasks. We tested our device with a 3.7V 500 mAh battery, measuring total power consumption (mW) and battery life (hours). The device operates continuously without sleep, consuming approximately 66.254 mW, resulting in a battery life of about 27.92 hours.

Table VII compares the proposed CNN model's size in terms of RAM (kB) and Flash (kB) using TFLM. It indicates that optimizing the model from 32-bit floating point to 8-bit integers reduced the Flash size from 96.512 kB to 43.654 kB and the RAM size from 10.325 kB to 8.647 kB. The size of the weights (kB) of the proposed non-quantized CNN model (floating-point model) after conversion to the embedded CNN (8-bit integer) using TensorFlow Lite with quantization has decreased from 94.54 kB to approximately 23.79 kB, which is about four times smaller.

F. Ablation Analysis

In Section IV-C, we utilized the TPE tuner, a SMBO algorithm, in conjunction with the NNI toolkit and 10-fold CV to

TABLE VIII

ARCHITECTURES OF THE TOP 5 MODELS WITH BEST PERFORMANCE AND THE TOP 5 MODELS WITH WORST PERFORMANCE AFTER HYPERPARAMETER TUNING

Top-Rank	Conv2d _1	Conv2d _2	Opt	Lr	Batch Size	ACC
Top-5 with highest accuracy	16	32	Adam	0.001	16	95.32%
	8	16	Adam	0.0001	16	93.78%
	16	16	Adam	0.01	4	93.68%
	32	16	SGD	0.001	16	93.53%
	16	16	Adam	0.001	16	93.29%
Top-5 with lowest accuracy	8	32	Adam	0.01	8	88.55%
	16	8	SGD	0.01	16	88.85%
	32	32	SGD	0.01	8	88.95%
	32	16	Adam	0.001	8	90.13%
	16	16	SGD	0.001	8	90.16%

TABLE IX

MULTI-CLASSIFICATION PERFORMANCE RESULTS OF CNN AND ANN WITH 10-FOLD CV AND LOOCV

Model	Strategy	Evaluation measures				
		Acc	Spe	Sen	Pre	F1
CNN	10-fold CV	0.9012 ±0.0177	0.9006 ±0.0326	0.9012 ±0.0175	0.9020 ±0.0168	0.9010 ±0.0184
	LOOCV	0.8621 ±0.0246	0.8601 ±0.0442	0.8621 ±0.0201	0.8633 ±0.0248	0.8618 ±0.0144
	10-fold CV	0.8728 ±0.0368	0.8733 ±0.0549	0.8728 ±0.0368	0.8754 ±0.0354	0.8725 ±0.0371
ANN	LOOCV	0.8185 ±0.0562	0.8201 ±0.0893	0.8185 ±0.0245	0.8236 ±0.0536	0.8167 ±0.0582

select and adjust the hyperparameters for the proposed CNN model. Our aim was to achieve the highest accuracy in stress detection by optimizing several parameters, including conv2d_1, conv2d_2, optimizer, learning rate, and batch size. As depicted in Fig. 10, the hyperparameter set (16, 32, Adam, 0.001, and 16) yielded the highest accuracy at 95.32%. The set (8, 16, Adam, 0.0001, and 16) followed closely with an accuracy of 93.78%, while the set (8, 32, Adam, 0.01, and 8) exhibited the lowest accuracy at 88.55%. Table VIII provides a detailed description of the architectures of the Top 5 models with the best performance and the Top 5 models with the worst performance following hyperparameter tuning.

In this study, we propose an on-chip BTE-EEG embedded system for early detection of mental stress. Initially, we used a two-class classification method to distinguish between relaxation and stress states. However, to evaluate our system's performance in a multi-class context, we expanded to a three-class model to differentiate non-stress, low-level stress, and high-level stress states, based on participants' self-assessment and NASA-TLX scores ranging from 0 to 10 for non-stress, 11 to 15 for low-level stress, and 16 to 20 for high-level stress. Data included 16,640 images for high-level stress, 15,760 for low-level stress, and 21,600 for non-stress. We evaluated the model using 10-fold CV and LOOCV techniques. Table IX shows that our CNN model consistently outperforms the ANN model, achieving higher accuracy, specificity, sensitivity, precision, and F1-scores. Specifically, the CNN model achieved 0.9012 ± 0.0177 accuracy

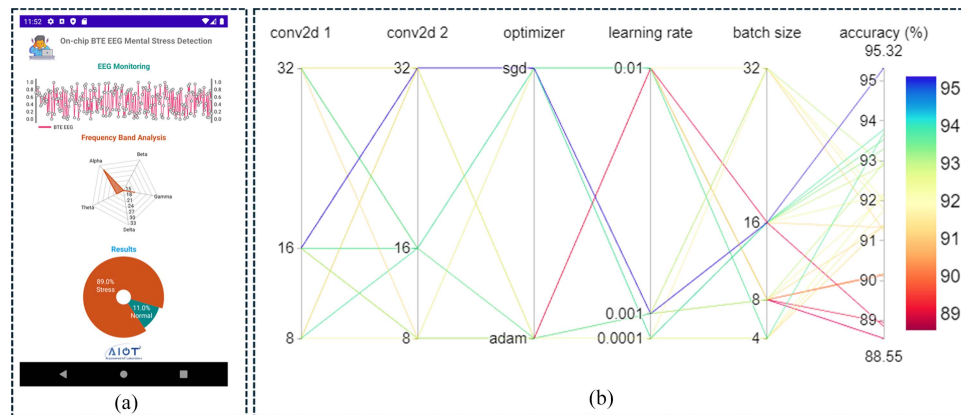


Fig. 10. (a) User Interface (UI) design for the smartphone application used for remote monitoring; (b) The results of hyperparameter tuning for the proposed CNN model.

with 10-fold CV and 0.8621 ± 0.0246 with LOOCV, compared to the ANN model's 0.8728 ± 0.0368 and 0.8185 ± 0.0562 , respectively.

G. Comparison With Previous State-Of-The-Art Studies

Numerous prior studies have explored the utilization of EEG signals for detecting mental stress. Most of these studies have concentrated on stress detection systems employing multiple sensor electrodes to gather EEG signals from the scalp [44], [45], [49]. However, these systems often become bulky, time-consuming to set up, and uncomfortable for prolonged wear, rendering them impractical for real-world applications. For example, A. Akella et al. [44] proposed a method using brain cortical EEG with a machine learning autoencoder to classify multi-level stress responses, achieving an accuracy of up to 91% using a NeuroScan EEG system, SynAmps 2 with 32 channels. Similarly, P. R. Fu et al. [45] combined symmetric convolutional and adversarial neural networks for mental stress classification using EEG signals collected from a Wireless LiveAmp 32 with 32 channels, achieving an accuracy of 87.62% for the classification of four stages.

To address the challenges associated with using multiple channels for EEG signal collection, some studies have sought to simplify setup and reduce device size by decreasing the number of signal collection locations. For instance, P. Arpaia et al. [46] proposed a wearable EEG instrument for real-time monitoring of worker stress with only 2 EEG electrodes (Fp1, Fp2 channels) from the prefrontal cortex, achieving an accuracy of up to 90% for 2-level stress analysis. Similarly, T. Y. Wen et al. [47] collected EEG signals under stress induced by virtual reality horror videos and intelligence quotient tests, achieving very high accuracy of up to 98% for 3-level stress analysis using only 2 measurement locations (Fp1, Fp2 channels). A. Arsalan et al. [48] used a commercially available four-channel Muse EEG headband with 4 measurement locations including AF7, AF8, TP9, and TP10 channels for the classification of perceived mental stress, achieving accuracies of 92.85% for 2-level stress classification and 64.28% for 3-level stress classification.

Despite the use of a limited number of EEG electrodes on the scalp, wearing them for an extended period can still cause discomfort to users. Therefore, ear EEG can be considered an optimal solution for this issue. However, there have been few studies conducted on using ear EEG signals for stress detection to date. One such study by Joong Hoon et al. [21] introduced a stress monitoring system using a multimodal bio-sensing headset for collecting in-ear EEG and ECG signals. This study demonstrated that using in-ear EEG with a CNN model achieved accuracies of 68% per frame and 75.8% per session for 2-level stress analysis. However, whether using standard scalp EEG or in-ear EEG, these studies share the commonality of performing data processing on PC hardware to execute detection algorithms based on data received from wearable devices. This approach results in high energy consumption, reduces real-time detection capability, and makes the system bulky and less user-friendly.

To address these limitations, we propose a novel solution by developing wearable devices with integrated on-chip stress detection capabilities. Our embedded device is designed for on-chip processing of stress detection with BTE EEG, providing a lightweight and affordable option. It includes on-chip noise removal, on-chip spectrogram conversion, and compact on-chip CNN model inference while ensuring comfort for long-term use and facilitating wireless data collection. Table X presents a detailed comparison of relevant information between our proposed system and previous state-of-the-art studies on stress systems.

This study involved a group of 15 individuals spanning various age groups who participated in our experiment. Each participant underwent 5 trials to ensure dataset diversity. This methodological approach facilitated the capture of diverse responses and behaviors, thereby enriching the depth of our data. By conducting multiple trials, we aimed to mitigate the impact of variability and random fluctuations inherent in single-trial studies, thus enhancing the reliability of our findings. Furthermore, the use of multiple trials per participant enabled us to assess the consistency and stability of our system's performance across different stress scenarios.

However, to enhance the robustness and generalizability of our findings to real-world scenarios, we recognize the necessity

TABLE X

COMPARISON OF RELEVANT INFORMATION BETWEEN OUR PROPOSED SYSTEM AND PREVIOUS STATE-OF-THE-ART STUDIES ON STRESS SYSTEMS

Study	No. of Subject	Dataset /Device	Stressors	Measurement Location	Methodology	Result	Processing
[46]	10	EEG-SMT by Olimex	High mental load and negative emotional conditions during the interaction with a cobot	Scalp EEG (Fp1, Fp2)	Time domain features with SVM, k-NN, random forest, and ANN	Accuracy: 90% for 2-level stress analysis	On-PC
[44]	80	NeuroScan EEG system, SynAmps 2	Preparing for a speech, delivering a speech, and performing an arithmetic task.	Scalp EEG (32 channels)	Latent features and SVM	Accuracy: 91% for multiple level stress analysis	On-PC
[47]	50	g.MOBilab+	The virtual reality horror video and intelligence quotient test	Scalp EEG (Fp1, Fp2)	The feature of Beta-band absolute power at right (Fp2) prefrontal region and SVM	Accuracy: 98% for 3-level stress analysis	On-PC
[49]	32	DEAP dataset and their own dataset	Under different conditions and facing various stressors on real construction sites.	Scalp EEG	Online Multitask Learning Algorithms (OMTL) with time domain feature, frequency domain feature and neuroanatomical	Accuracy: 71.14% on the DEAP dataset Accuracy: 77.61% on their own dataset	On-PC
[48]	28	Interaxon Muse headband	Public speaking and post activity	Scalp EEG (AF7, AF8, TP9 TP10)	Five feature groups from theta band with feature selection algorithm	Accuracy: 92.85% for 2-level stress classification Accuracy: 64.28% for 3-level stress classification	On-PC
[45]	22	Wireless LiveAmp 32	Trier Social Stress Test	Scalp EEG (32 channels)	SDCAN and EA-SDCAN for stress classification	Accuracy: 87.62% on the classification of four stages Accuracy: 81.45% on the classification of five stages	On-PC
[21]	10	Custom-designed device	Stroop, memory search, and mental arithmetic	In-Ear EEG (2 channels)	CNN model with raw time-domain EEG signals	Accuracy: 68% for 2-level stress analysis per frame Accuracy: 75.8% for 2-level stress analysis per session	On-PC
Ours	15	Custom-designed device	Stroop, mental arithmetic	BTE EEG (single channel)	End-to-end on-chip mental stress detection using on-chip signal-to-spectrogram conversion and an embedded tiny CNN model	Accuracy: 95.32% with 10-fold CV for 2-level stress analysis Accuracy: 91.72% with LOOCV for 2-level stress analysis	On-Chip

to expand our experimental scope. Future research endeavors will involve increasing both the number of participants and trials conducted. This expansion aims to achieve broader demographic representation and facilitate a more comprehensive data analysis. Increasing the sample size will serve to validate and generalize our findings effectively, thereby aligning them more closely with benchmarks established by larger-scale studies in the field.

In this study, we used the FFT to convert EEG signals into spectrograms. While other methods like the CWT or SLT offer better time localization, FFT is the fastest and most efficient for signal processing [33]. Its speed made it the preferred choice for converting EEG signals to spectrograms on-chip. Previous research has shown that FFT enables our system to perform well and is suitable for real-time applications due to its rapid processing capabilities. We used FFT-based spectrogram images as inputs for our proposed CNN model, which is designed for image-based classification. A key feature of our embedded CNN model is its adaptation for microcontrollers through

min-max quantization. This process reduces the model size and improves hardware latency by converting weights from 32-bit floating-point precision to 8-bit integer precision. Although this quantization slightly decreased the accuracy of stress classification by approximately 0.76% (from 94.32% to 93.56%), it significantly improved inference speed, reducing it from 36.54 ms to 31.79 ms. Despite the minor accuracy reduction, the combination of FFT for on-chip EEG signal conversion and the quantized embedded CNN model demonstrated promising results in both accuracy and processing speed. Future research will focus on exploring advanced feature extraction techniques and more effective classification models for stress detection.

V. CONCLUSION

The study proposed a novel approach for efficient mental stress detection by leveraging EEG analysis and on-chip neural networks. Through the integration of behind-the-ear EEG signals and embedded CNN models, our system offered a novel

solution with significant potential for real-world applications. The development of a wearable embedded device facilitates seamless EEG signal capture, while advanced signal processing techniques and frequency band analysis contribute to accurate stress level classification. Our experimental results demonstrate the high performance of the proposed system, underscoring its effectiveness in stress detection tasks. The integration of BTE EEG analysis and on-chip CNNs represents a notable advancement in medical assistance tools, with implications for personalized healthcare and well-being monitoring. Moving forward, further research and refinement of this approach could lead to enhanced stress management strategies and improved patient outcomes in clinical settings.

In this study, we assessed our proposed stress detection system using data from 15 subjects, each participating in 5 trials. The evaluation was conducted using five metrics in both 10-fold CV and LOOCV. In the LOOCV approach, each subject's data served as the testing set while the data from the remaining subjects were used for training, ensuring that the model was evaluated comprehensively across different individuals and enhancing its practical applicability. Although our system demonstrated high performance in various evaluation scenarios, additional research is necessary to validate its effectiveness in real-world conditions. Future work will involve collecting and analyzing data from a larger number of subjects and conducting more experimental trials to provide sufficient data for robust model training and assessment. Moreover, it is essential to evaluate the model under diverse real-world conditions, taking into account variations in age, gender, health status, and other individual factors to ensure its broad applicability and reliability.

REFERENCES

- [1] S. A. Hosseini, M. A. Khalilzadeh, and S. Changiz, "Emotional stress recognition system for affective computing based on bio-signals," *Biol. Syst.*, vol. 18, pp. 101–114, 2010.
- [2] F. Al-Shargie, M. Kiguchi, N. Badruddin, S. C. Dass, A. F. M. Hani, and T. B. Tang, "Mental stress assessment using simultaneous measurement of EEG and fNIRS," *Biomed. Opt. Exp.*, vol. 7, no. 10, pp. 3882–3898, 2016.
- [3] M. Irie, S. Asami, S. Nagata, M. Miyata, and H. Kasai, "Relationships between perceived workload, stress and oxidative DNA damage," *Int. Arch. Occup. Environ. Health*, vol. 74, no. 2, pp. 153–157, 2001.
- [4] C. Schubert, M. Lambert, R. Nelesen, W. Bardwell, J.-B. Choi, and J. Dimsdale, "Effects of stress on heart rate complexity—A comparison between short-term and chronic stress," *Biol. Psychol.*, vol. 80, no. 3, pp. 325–332, 2009.
- [5] S. Cohen, R. C. Kessler, and L. U. Gordon, *Measuring Stress: A Guide For Health and Social Scientists*. Oxford, U.K.: Oxford Univ. Press on Demand, 1997.
- [6] D. H. Hellhammer, S. Wüst, and B. M. Kudielka, "Salivary cortisol as a biomarker in stress research," *Psychoneuroendocrinology*, vol. 34, no. 2, pp. 163–171, 2009.
- [7] G. K. Verma and U. S. Tiwary, "Multimodal fusion framework: A multi-resolution approach for emotion classification and recognition from physiological signals," *NeuroImage*, vol. 102, pp. 162–172, 2014.
- [8] J. D. Bremner et al., "MRI-based measurement of hippocampal volume in patients with combat-related posttraumatic stress disorder," *Amer. J. Psychiatry*, vol. 152, no. 7, 1995, Art. no. 973.
- [9] S. L. Rauch et al., "Exaggerated amygdala response to masked facial stimuli in posttraumatic stress disorder: A functional MRI study," *Biol. Psychiatry*, vol. 47, no. 9, pp. 769–776, 2000.
- [10] F. Al-Shargie, T. B. Tang, and M. Kiguchi, "Assessment of mental stress effects on prefrontal cortical activities using canonical correlation analysis: An fNIRS-EEG study," *Biomed. Opt. Exp.*, vol. 8, no. 5, pp. 2583–2598, May 2017.
- [11] J. D. Bremner et al., "MRI and PET study of deficits in hippocampal structure and function in women with childhood sexual abuse and posttraumatic stress disorder," *Amer. J. Psychiatry*, vol. 160, no. 5, pp. 924–932, 2003.
- [12] J. Wang et al., "Perfusion functional MRI reveals cerebral blood flow pattern under psychological stress," *Proc. Natl. Acad. Sci. USA*, vol. 102, no. 49, pp. 17804–17809, 2005.
- [13] K. Dedovic, C. D'Aguiar, and J. C. Pruessner, "What stress does to your brain: A review of neuroimaging studies," *Can. J. Psychiatry*, vol. 54, no. 1, pp. 6–15, 2009.
- [14] X. Hou, Y. Liu, O. Sourina, Y. R. E. Tan, L. Wang, and W. Mueller-Wittig, "EEG based stress monitoring," in *Proc. IEEE Syst. Man Cybern.*, 2015, pp. 3110–3115.
- [15] G. Matthews, *Human Performance: Cognition Stress and Individual Differences*. Hove, U.K.: Psychology Press, 2000.
- [16] N. Sharma and T. Gedeon, "Objective measures sensors and computational techniques for stress recognition and classification: A survey," *Comput. Methods Programs Biomed.*, vol. 108, no. 3, pp. 1287–1301, 2012.
- [17] S. Ranabir and K. Reetu, "Stress and hormones," *Indian J. Endocrinol. Metab.*, vol. 15, no. 1, pp. 18–22, 2011.
- [18] E. Alyan, S. Arnau, J. E. Reiser, S. Getzmann, M. Karthaus, and E. Wascher, "Decoding eye blink and related EEG activity in realistic working environments," *IEEE J. Biomed. Health Inform.*, vol. 27, no. 12, pp. 5745–5754, Dec. 2023.
- [19] H.-T. Nguyen, N.-D. Mai, B. Giin Lee, and W.-. Chung, "Behind-the-ear EEG-based wearable driver drowsiness detection system using embedded tiny neural networks," *IEEE Sensors J.*, vol. 23, no. 19, pp. 23875–23892, Oct. 2023.
- [20] P. Kidmose, D. Looney, M. Ungstrup, M. L. Rank, and D. P. Mandic, "A study of evoked potentials from ear-EEG," *IEEE Trans. Biomed. Eng.*, vol. 60, no. 10, pp. 2824–2830, Oct. 2013.
- [21] J. H. Lee et al., "Stress monitoring using multimodal Bio-sensing headset," in *Proc. Extended Abstr. 2020 CHI Conf. Hum. Factors Comput. Syst.*, New York, NY, USA, 2020, pp. 1–7.
- [22] M. Bhagubai et al., "The power of ECG in semi-automated seizure detection in addition to two channel behind-the-ear EEG," *Bioengineering*, vol. 10, no. 4, 2023, Art. no. 491.
- [23] G. Li, B.-L. Lee, and W.-Y. Chung, "Smartwatch-based wearable EEG system for driver drowsiness detection," *IEEE Sensors J.*, vol. 15, no. 12, pp. 7169–7180, Dec. 2015.
- [24] G. Li and W.-Y. Chung, "Combined EEG-gyroscope-tDCS brain machine interface system for early management of driver drowsiness," *IEEE Trans. Hum.-Mach. Syst.*, vol. 48, no. 1, pp. 50–62, Feb. 2018.
- [25] N. S. Karuppusamy and B.-Y. Kang, "Multimodal system to detect driver fatigue using EEG gyroscope and image processing," *IEEE Access*, vol. 8, pp. 129645–129667, 2020.
- [26] P. Warden and D. Situnayake, and ML Tiny, *Machine Learning with Tensorflow Lite On Arduino and Ultra-Low-Power Microcontrollers*, Sebastopol, CA, USA: O'Reilly Media, 2019.
- [27] S. Chaabene, B. Bouaziz, A. Boudaya, A. Hökelmann, A. Ammar, and L. Chaari, "Convolutional neural network for drowsiness detection using EEG signals," *Sensors*, vol. 21, no. 5, Mar. 2021, Art. no. 1734.
- [28] S. Gangadharan K and A. P. Vinod, "Drowsiness detection using portable wireless EEG," *Comput. Methods Programs Biomed.*, vol. 214, Feb. 2022, Art. no. 106535.
- [29] N. Pham et al., "Detection of microsleep events with a behind-the-ear wearable system," *IEEE Trans. Mobile Comput.*, vol. 22, no. 2, pp. 841–857, Feb. 2023, doi: [10.1109/TMC.2021.3090829](https://doi.org/10.1109/TMC.2021.3090829).
- [30] S. Debener et al., "Unobtrusive ambulatory EEG using a smartphone and flexible printed electrodes around the ear," *Sci Rep.*, vol. 5, 2015, Art. no. 16743.
- [31] K. Wang, H. Wen, and G. Li, "Accurate frequency estimation by using three-point interpolated discrete Fourier transform based on rectangular window," *IEEE Trans. Ind. Inform.*, vol. 17, no. 1, pp. 73–81, Jan. 2021.
- [32] S. H. Mirfarshbafan, S. Taner, and C. Studer, "SMUL-FFT: A streaming multiplierless fast fourier transform," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 68, no. 5, pp. 1715–1719, May 2021.
- [33] V. V. Moca, H. Bärzan, A. Nagy-Dăbăcan, and R. C. Mureşan, "Time-frequency super-resolution with superlets," *Nat. Commun.*, vol. 12, no. 1, 2021, Art. no. 337.
- [34] Z. Li, Fan Liu, W. Yang, S. Peng, and J. Zhou, "A survey of convolutional neural networks: Analysis, applications, and prospects," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 6999–7019, Dec. 2022.

- [35] C. Janiesch, P. Zschech, and K. Heinrich, "Machine learning and deep learning," *Electron. Markets*, vol. 31, no. 3, pp. 685–695, 2021.
- [36] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: visual explanations from deep networks via gradient-based localization" *Int. J. Computer Vision*, vol. 128, pp. 336–359, 2016.
- [37] B. Jacob et al., "Quantization and training of neural networks for efficient integer-arithmetic-only inference," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 2704–2713.
- [38] M. G. Bleichner and S. Debener, "Independent component decomposition of around ear EEG data to detect artifacts," in *Proc. IEEE Int. Conf. Syst. Man Cybern.*, Oct. 2019, pp. 3631–3634.
- [39] F. Al-Shargie, M. Kiguchi, N. Badruddin, S. C. Dass, A. F. Mohammad Hani, and T. Boon Tang, "Mental stress assessment using simultaneous measurement of EEG and fNIRS," *Biomed. Opt. Exp.*, vol. 7, no. 10, pp. 3882–3898, 2016.
- [40] J. H. Lee et al., "Stress monitoring using multimodal bio-sensing headset," in *Proc. Extended Abstr. 2020 CHI Conf. Hum. Factors Comput. Syst.*, 2020, pp. 1–7.
- [41] K. Dedovic, R. Renwick, N. Khalili Mahani, V. Engert, S. J. Lupien, and J. C. Pruessner, "The Montreal imaging stress task: Using functional imaging to investigate the effects of perceiving and processing psychosocial stress in the human brain," *J. Psychiatry Neurosci.*, vol. 30, no. 5, pp. 319–325, 2005.
- [42] S. G. Hart and L. E. Staveland, "Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research," *Adv. Psychol.*, vol. 52, pp. 139–183, 1988.
- [43] I. Gridin, "Introduction to neural network intelligence," in *Automated Deep Learning Using Neural Network Intelligence: Develop and Design Pytorch and TensorFlow Models Using Python*, Berkeley, CA, USA: A Press, 2022, pp. 1–30.
- [44] A. Akella et al., "Classifying multi-level stress responses from brain cortical EEG in nurses and non-health professionals using machine learning auto encoder," *IEEE J. Transl. Eng. Health Med.*, vol. 9, 2021, Art. no. 2200109.
- [45] R. Fu et al., "Symmetric convolutional and adversarial neural network enables improved mental stress classification from EEG," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 30, pp. 1384–1400, 2022.
- [46] P. Arpaia, N. Moccaldi, R. Prevete, I. Sannino, and A. Tedesco, "A wearable EEG instrument for real-time frontal asymmetry monitoring in worker stress analysis," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 10, pp. 8335–8343, Oct. 2020.
- [47] T. Y. Wen and S. A. M. Aris, "Hybrid approach of EEG stress level classification using K-means clustering and support vector machine," *IEEE Access*, vol. 10, pp. 18370–18379, 2022.
- [48] A. Arsalan, M. Majid, A. R. Butt, and S. M. Anwar, "Classification of perceived mental stress using a commercially available EEG headband," *IEEE J. Biomed. Health Inform.*, vol. 23, no. 6, pp. 2257–2264, Nov. 2019.
- [49] H. Jebelli, M. Mahdi Khalili, and S. Lee, "A continuously updated, computationally efficient stress recognition framework using electroencephalogram (EEG) by applying online multitask learning algorithms (OMTL)," *IEEE J. Biomed. Health Inform.*, vol. 23, no. 5, pp. 1928–1939, Sep. 2019.